

6.5.5 Momentum (HT only)

6.5.5.1 Momentum is a property of moving objects

Content	Key opportunities for skills development
Momentum is defined by the equation: $\text{momentum} = \text{mass} \times \text{velocity}$ $p = m v$ momentum, p, in kilograms metre per second, kg m/s mass, m, in kilograms, kg velocity, v, in metres per second, m/s	WS 1.2 MS 3b, c Students should be able to recall and apply this equation.

6.5.5.2 Conservation of momentum

Content	Key opportunities for skills development
In a closed system, the total momentum before an event is equal to the total momentum after the event. This is called conservation of momentum. Students should be able to use the concept of momentum as a model to describe and explain examples of momentum in an event, such as a collision.	AT 1, 2, 3 Investigate collisions between laboratory trolleys using light gates, data loggers or ticker timers to measure and record data.